

Short Questions

1. What do you mean by Entity-Relationship Diagram? Explain

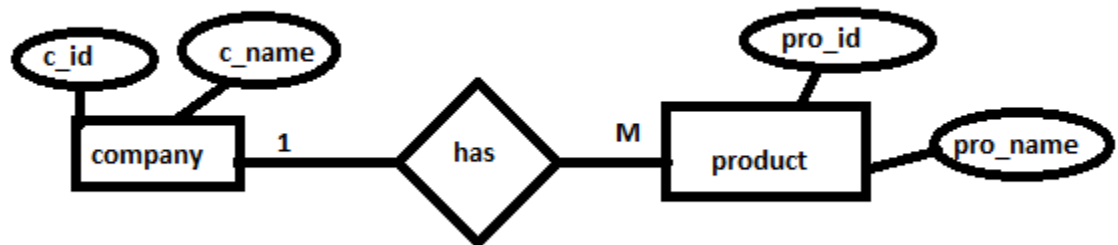
An Entity-Relationship Diagram (ERD) is a visual representation used in database design to illustrate the logical structure of a database system.

Entity:An entity in an ER Model is a real-world entity like thing, person, place, unit, object or any item having properties called attributes.

Attributes:An attribute defines the information about the entity that needs to be stored.

relationship:The logical association among entities is called relationship.

For example:In a shop database, a **company and product** are considered as entities ,whose c_id,c_name,pro_id etc. is considered as an attribute.



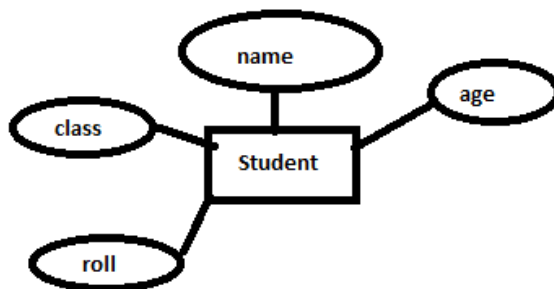
2. Define the term entity and give an example.

An entity in an ER Model is a real-world entity like thing, person, place, unit, object or any item having properties called attributes.

For example:

In a school database,

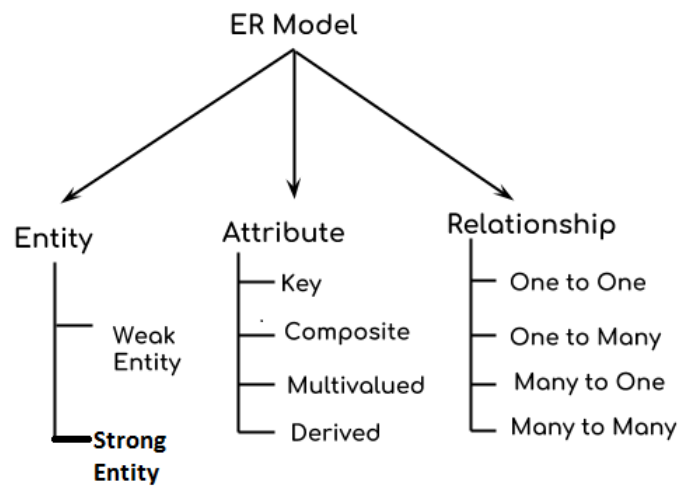
a **student** is considered as an entity ,whose name, age, class, etc. is considered as an attributes



3. What are the components of the ER-model?

The E-R diagram has three main components. 1) Entity 2) Attribute 3) Relationship

Components of a ER Diagram



Components of ER Diagram

4. Define the attribute and its type with suitable examples.

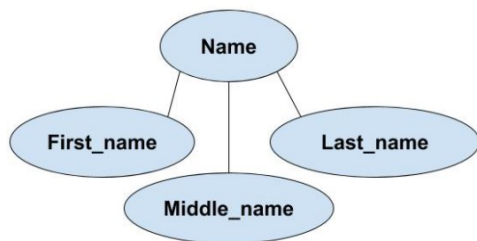
An attribute defines the information about the entity that needs to be stored.

Types of attributes:

- a. Simple and Composite Attributes

The composite attribute is an attribute that can be subdivided into other single attributes with meanings of their own.

For example:



A simple attribute is just an attribute that cannot be subdivided into parts.

For example:



- b. Single-valued and multi-valued Attributes

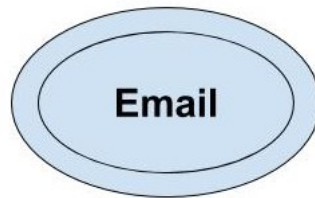
The single-value attribute can only have one value,

For example, Section of a Student.



while the multi-valued attributes usually can store multiple data in them.

For example, A student can have more than one e-mail address.



c. Stored attributes and Derived Attributes

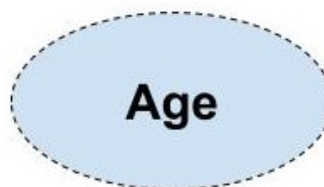
Stored attributes are stored in a database and can be used to find derived attributes.

Example: Date of Birth(DOB) of Student.



Derived attributes are not stored in the database. It is calculated from stored attributes.

For example, The value of age attribute is derived from the DOB(date of birth) attribute.



d. Keys and non-keys Attributes

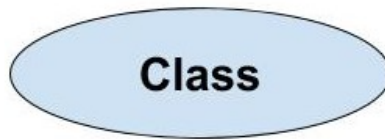
A key attribute is used to uniquely identify the entities of an entity type

For example:



All the other attributes other than the key attribute are the non-key attributes.

Example: Class, Section, Age, Name etc, are the non-key attributes.



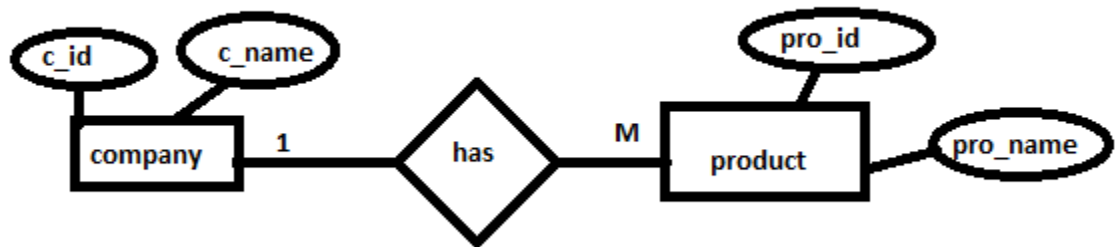
5. What is the derived attribute? Explain it with an appropriate example.

6. Define relationships and give an example.

The logical association among entities is called relationship.

For example: In a shop database,

a **company and product** are considered as entities, whose c_id, c_name, pro_id etc. is considered as an attribute.



7. Define the degree of relationship.

The degree of a relationship in an Entity-Relationship Diagram (ERD) represents the number of entity sets involved in that relationship.

8. Explain the major features of the primary key.

- i. Unique
- ii. Not NULL

Long Questions

1. What is key? Explain the major features of the primary key.

Keys are crucial elements in a Relational Database Management System (RDBMS). They are used to establish and maintain relationships between different tables within the database.

The Primary Key is a specific type of key that holds significant importance. It's a column in a table that uniquely identifies each row or record in that table. The major features of a primary key include:

- **Uniqueness:** No two rows in the table can have the same value for the primary key. This ensures that each record can be distinguished from others.
- **Non-Repeatability:** The primary key value cannot be duplicated within the same column. This guarantees the uniqueness of each record.
- **Non-Nullability:** A primary key value cannot be null (empty). This ensures that every record has a valid and distinct identifier.
- **Stability:** Ideally, the value of a primary key should remain constant over time to maintain integrity.

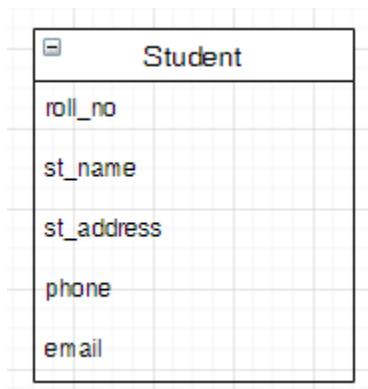
For example: In a student table, the column "roll_no" is the primary key. It uniquely identifies each student and guarantees that no two students have the same roll number.

2. What is the key or Database Keys in the Database system? List out the different keys used in Database.

Keys are a very important part of a Relational database. They are used to establish and identify relations between tables.

1. **Primary Key:** column in a table that can never be repeated is called primary key.
For example:
roll_no is a primary key
2. **Super Key:** The primary key itself or primary key with any other attributes that helps to uniquely identify tuples in a table is called super key

In student table,



Student
roll_no
st_name
st_address
phone
email

For example:

roll_no+st_name+st_address is a super key
roll_no is also a super key

1. **Candidate keys:** The primary key itself or the primary key with other suitable attributes which can be possible primary keys that uniquely identify tuples in a table is called candidate keys.

For example:

roll_no+phone+email is a candidate key

4. Foreign key

Foreign keys are the column of the table which is used to point to the primary key of another table.

5. **Composite key**: More than one Foreign key in a table that uniquely identifies every row in

Composite Key
↑

student_id	subject_id	marks	exam_name

the table.

6. Secondary or Alternative key

The candidate keys which are not selected as primary keys are known as secondary keys or alternative keys.

For example:

phone+email is a secondary key

Phone is also a secondary key

3. Explain the differences between a weak and a strong entity set.

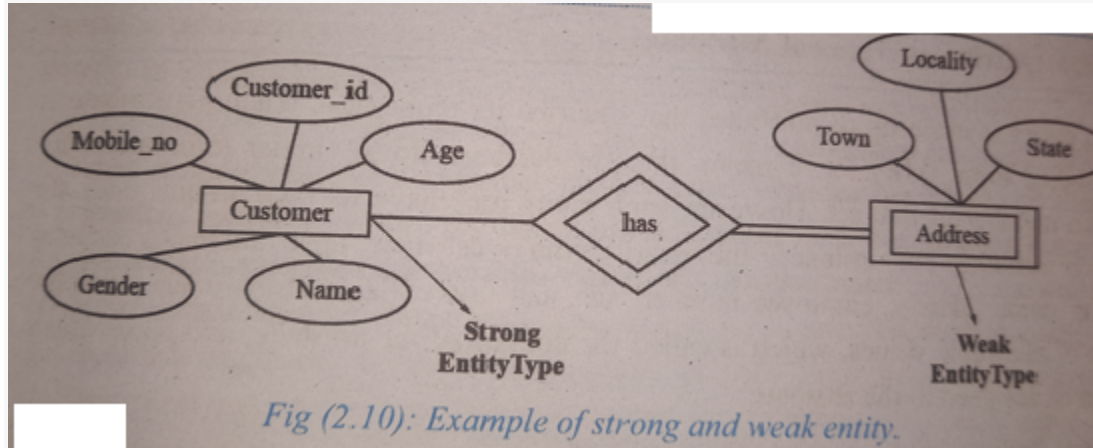
Strong Entity Type:

- A strong entity type is a group of entities that have a key attribute, allowing each entity to be uniquely identified.
- Strong entities can exist independently, meaning they don't rely on any other entity for identification.
- They are represented using a single outlined rectangle in an ER diagram.
- For example, in a "Student" table, the "Student" entity is strong because it has a key attribute, such as "Roll no," that can uniquely identify each student.

Weak Entity Type:

- A weak entity type lacks a key attribute and cannot be identified on its own. It relies on a related strong entity for unique identification.
- Weak entities exist in the context of a strong entity, and they have a partial key used in conjunction with the strong entity's key.
- They are represented using a double outlined rectangle in an ER diagram.
- The relationship between a weak entity type and a strong entity type is called an "identifying relationship," represented by a double outlined diamond in the ER diagram.
- For instance, in the parent-child relationship, a "Child" entity could be weak, relying on the "Parent" entity for identification.

Example: Suppose you have a "Customer" table and an "Address" table. The "Address" table could be a weak entity type because it lacks a unique key on its own. It relies on a strong entity, "Customer," for identification. The "Customer_id" attribute from the "Customer" table serves as a key to uniquely identify addresses. This demonstrates the concept of a weak entity type relying on a strong entity for its existence.



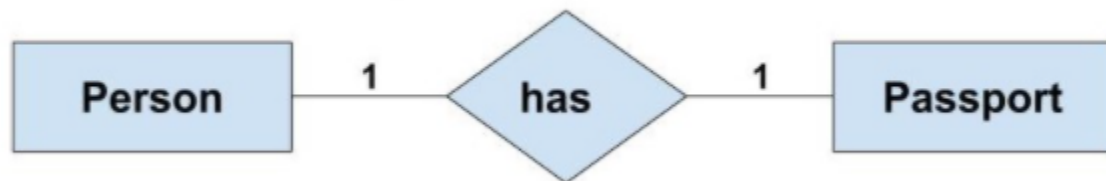
4. Explain different types of relationships with examples.

The logical association among entities is called relationship.

Types of relationship are:

a). one to one

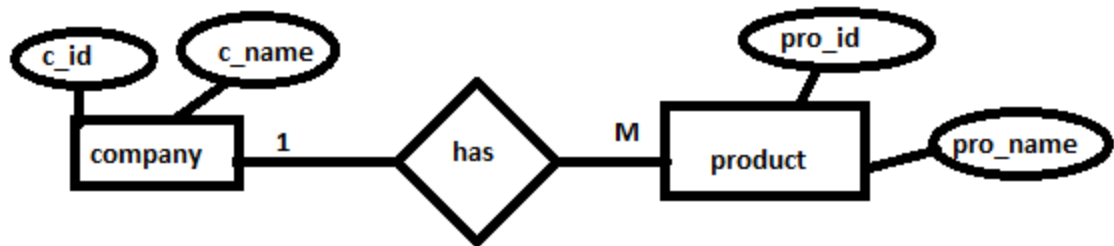
One entity can be associated with at most one entity and vice versa.



b). one to many

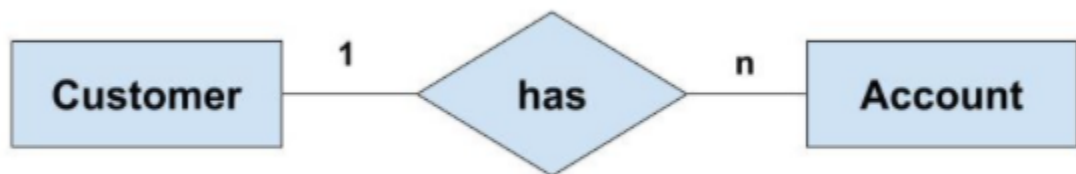
One entity can be associated with more than one entities however an entity from other side can be associated with at most one entity.

For example:



c). many to one

More than one entities can be associated with at most one entity, however an entity can be associated with more than one entity from the other side.

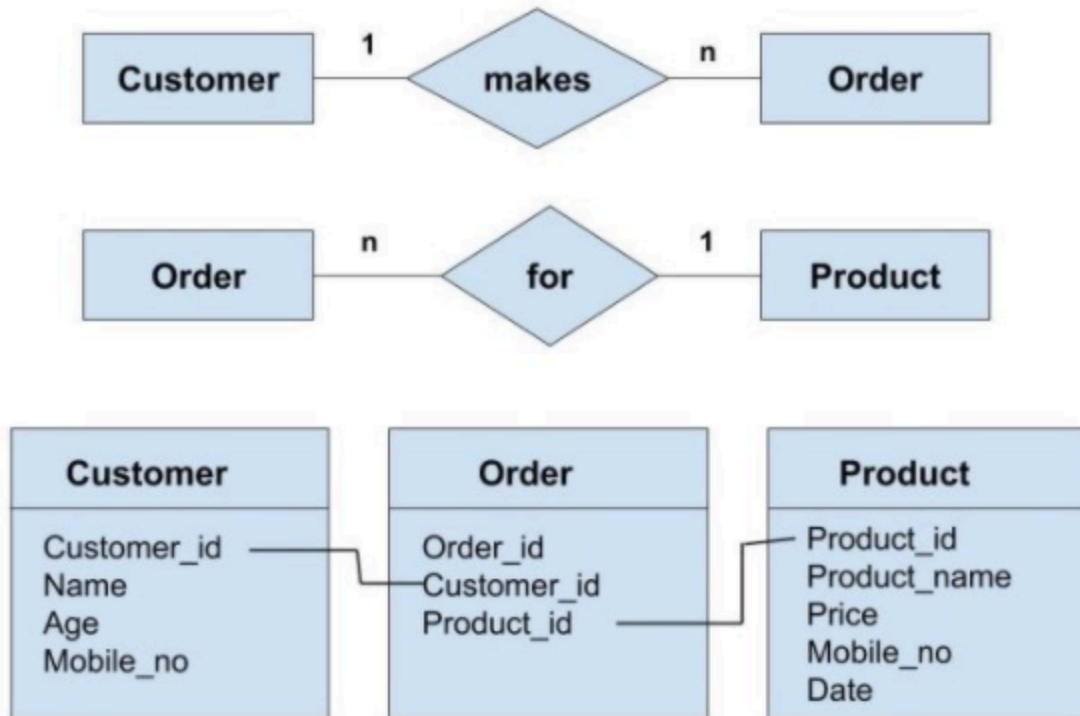


d). many to many

One entity can be associated with more than one entity and vice versa.



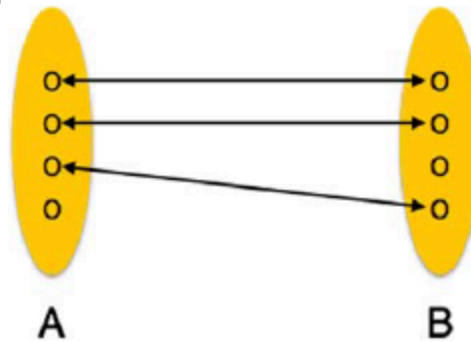
If there are many to many relationships between entities then we need a third linking entity which links two entities.



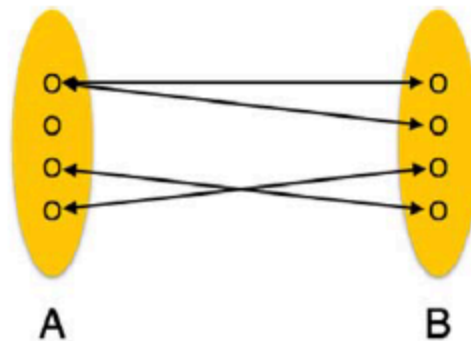
5. What is mapping cardinalities? Discuss with an example

Cardinality defines the number of entities in one entity set, which can be associated with the number of entities of other set via relationship set.

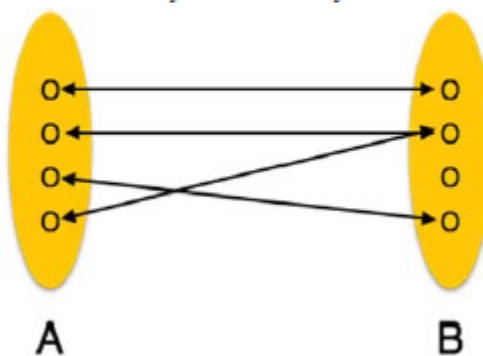
- **One-to-one** – One entity from entity set A can be associated with at most one entity of entity set B and vice versa.



- **One-to-many** – One entity from entity set A can be associated with more than one entities of entity set B however an entity from entity set B, can be associated with at most one entity.



- **Many-to-one** – More than one entities from entity set A can be associated with at most one entity of entity set B, however an entity from entity set B can be associated with more than one entity from entity set A.



- **Many-to-many** – One entity from A can be associated with more than one entity from B and vice versa.

